

6

Analysis of Variance

CHAPTER OBJECTIVES

- Understand the nature and logic of the analysis of variance (ANOVA) from first principles.
- Conceptualize t -tests for means as a special case of the ANOVA model.
- Understand why performing multiple t -tests does not guard against inflating error rates.
- Evaluate assumptions in ANOVA, including normality and homogeneity of variance.
- Conduct one-way and two-way ANOVA in Python and interpret effects.
- Understand the concept of an interaction in factorial ANOVA and why science is mostly about interactions, not main effects.
- How to read an interaction graph in two-way ANOVA.
- Understand the difference between between-subjects ANOVA and repeated measures ANOVA.

The analysis of variance, or “ANOVA” for short, plays a central role in statistical modeling. Not only is it one of the most commonly applied statistical procedures across virtually all of the sciences, but learning ANOVA helps the student of statistics or science better understand the principles behind not only the ANOVA model, but of many other statistical models as well, even relatively advanced ones. In this day and age of high-speed computing, “big data,” and all the rest of it, classic ANOVA is sometimes perceived as a “basic” method that is somewhat “antiquated.” Nothing could be further from the truth. ANOVA, along with regression of the following chapter, are the backbone statistical methodologies of modern statistics, and one could easily give a full course in ANOVA and still have many topics still left to cover. Our treatment of the method here is quite brief. For the student interested in a much deeper study of the procedure and to learn just how technical ANOVA can get, he/she is encouraged to consult Scheffé (1999) for a thorough, if not challenging, treatment.

6.1 T-Tests for Means as a “Special Case” of ANOVA

There are two ways of conceptualizing ANOVA. The first is to perceive ANOVA models as extensions of t -tests for means where instead of only two groups, we now have three or more populations that we wish to compare. The second is to see the ANOVA model as a wider, more complete model and as a special case of the regression model. The second approach provides much more perspective into statistics than the first, but since we have not covered regression yet, our focus will be on understanding ANOVA as an extension of the t -test for means. Once we study regression in the chapter to follow, we will correctly place the ANOVA model in its proper context as a special case of the more **general linear model**.

Recall that in an independent samples t -test, we were interested in evaluating a null hypothesis about population means. In the ideal and prototypical circumstance, we had both a **control group** and a **treatment group**, and our null was a statement that the two samples arose from the same population. Our statistical alternative hypothesis was that they were selected from distinct populations. We expressed this null hypothesis symbolically with the following:

$$H_0 : \mu_1 = \mu_2$$

where μ_1 was the mean for the first population and μ_2 was the mean for the second population. The alternative hypothesis was that the population means were unequal. That inequality could take the form of μ_1 being greater than μ_2 or μ_2 being greater than μ_1 . Hence, we could, if we really wanted to, specify the alternative hypothesis as $H_1 : \mu_1 > \mu_2$ or $H_1 : \mu_1 < \mu_2$. Each of these were known as **one-sided alternatives**. However, because we were interested in potential differences in either direction, we instead typically set our alternative hypothesis as

$$H_1 : \mu_1 \neq \mu_2$$

This was known as a **two-sided alternative** hypothesis and allowed for either population mean to be greater than the other. The two-sided alternative merely hypothesized a difference in means, not direction. As we discussed when doing t -tests, it is generally the preferred alternative hypothesis to state as a competitor to the null. In the case of an experiment, you may think and hope the treatment mean will be larger (or smaller, depending on the context of the experiment) than the control mean, but should the opposite effect occur, surely you would be just as interested in rejecting the null. The two-sided alternative allows for a rejection in either case and hence is the preferred statement. Unless you have a very specific reason for performing a one-tailed test, you should virtually always be conducting two-sided ones.

From there, we constructed the t -test:

$$t = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The numerator in the independent samples t -test measured the difference between sample means. The extent to which the difference was large, all else equal, was the extent to which t would come out to be large as well. However, the mean difference

$\bar{y}_1 - \bar{y}_2$ was not the only thing going on in the t -test. The denominator also played a critical role and represented the estimated standard error of the difference in means. In the case where sample sizes are unequal, such that $n_1 \neq n_2$, we can pool variances, which yields the following t -test:

$$t = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

The pooled variance term s_p^2 , unpacked, is equal to $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$ and is essentially a sort of average of the two variances, s_1^2 and s_2^2 .

If you understand the logic of the t -test, you will have virtually no problems understanding ANOVA. But what is the logic of the t -test, exactly? We have unpacked the t -test in our earlier discussions including in that of statistical power, where we had to understand the components of the t -test in some detail in order to make sense of what gave a test more power than not. Recall that the t -test is comparing a mean difference in the numerator to an estimated standard error of the difference in the denominator. That is, the t -test is comparing an observed sample difference to variation we would expect, on average, under the null hypothesis. We have already mentioned that if the numerator were large, all else being equal, the value of t would likewise be large. However, what of the denominator? Notice that all else equal, if the denominator is large, then the value of t will be small. This makes sense, because since the denominator reflects the average variability one might expect to find under the null hypothesis, we see the t -test as a comparison of **observed to expected average difference or variation**. Recall in the chi-squared goodness-of-fit test that we compared observed to expected frequencies. Though the t -test deals with different data, the analogy is powerful in terms of understanding what the statistical test actually “does.” Hence, even if $\bar{y}_1 - \bar{y}_2$ is large, if the denominator is quite large, then t will not come out very large, since the difference $\bar{y}_1 - \bar{y}_2$ is not that “impressive” when compared to the denominator, $\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$. This is the logic of the t -test, to compare an observed difference to an average difference (in terms of variability) expected under the null.

So what does any of this have to do with ANOVA? Everything! As we will see, ANOVA works in an incredibly similar way. We will see that ANOVA effectively compares a numerator to a denominator, where the numerator also reflects a difference in means, only now, we typically have more than simply two means. The denominator will also reflect more than variabilities within only two groups. Hence, we will see that t -tests and ANOVA are very similar conceptually in what they accomplish. However, why do ANOVA in the first place? Why not simply perform multiple t -tests on our group of means to determine where mean differences lie? There is a very good reason for this, and hence before considering the ANOVA model in detail, we first need to understand its justification and motivation. That is, we need to understand better why performing multiple t -tests on a set of means is typically not a good idea. We will then introduce the ANOVA model as a solution to this problem.

6.2 Why Not Do Several t -Tests?

We have said that ANOVA is used when there are more than two means to compare. Well, as we have suggested, the obvious question that arises is why not simply do multiple t -tests? For example, a null hypothesis for a four-population problem would look like this:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

If we would like to compare means, why not simply evaluate all pairwise mean comparisons via independent samples t -tests? That is, why not evaluate null hypotheses such as $H_0 : \mu_1 = \mu_2$, $H_0 : \mu_1 = \mu_3$, and so on, for all pairwise comparisons via independent samples t -tests? At first glance, it would seem like an attractive option and save us the trouble of having to learn ANOVA! However, the problem is that, as you will recall, with each mean comparison using a t -test comes with it a type I error rate, which is equal to the significance level for the test, often set by convention at 0.05. When we perform several mean comparisons on the same data, this error rate multiplies and hence gets larger with the number of comparisons we are doing. How large it gets is beyond the scope of this book (for details, see Denis, 2021). However, as a very rough approximation, especially for a relatively small number of comparisons, one can think of the error rate as multiplying by the number of comparisons made. This is only an approximation, but helpful in terms of conceptualizing the inflating error rate.

Hence, if we made two comparisons, we would have approximately double the error rate, and so our error rate across both comparisons would no longer be 0.05 as it is for each comparison, but rather near 0.10 across both comparisons. Again, for reasons beyond this book, the error rate is not precisely additive like this, but it does not hurt to think of it in this way, especially when performing relatively few comparisons. The morale of the story is that **when you perform several comparisons on the same data, the theoretical type I error rate gets larger the more comparisons you perform**. Now, one could simply attempt to control the otherwise inflating error rate by using what is known as a **Bonferroni adjustment**, where alpha is split up by the number of comparisons one wishes to perform. For instance, if we are performing three mean comparisons, we could conduct each comparison at $0.05/3 = 0.0167$. This would help keep the overall type I error rate down, though at the expense of a loss of statistical power for each comparison. As you can imagine, as the number of means grows in a given problem, performing such corrections on alpha becomes implausible, since the power for any of the given comparisons will be quite low. For instance, for even a six-comparison problem, each comparison would be conducted at 0.0083 (i.e. $0.05/6$), which yields hardly any statistical power for each comparison. That is, rejecting the null hypothesis at a significance level of 0.0083 will be quite difficult since alpha is so small. As we will see, performing an ANOVA will be much more efficient.

Hence, we now have the motivation for performing ANOVA, to help control the inflating error rate. ANOVA will allow us to evaluate the null hypothesis while keeping the type I error rate at a nominal level. That is, ANOVA will allow us to learn whether there are mean differences, all the while keeping the type I error more or less at the level of 0.05 for the family of comparisons. In this sense, we can test an **omnibus null hypothesis** without allowing the error rate to inflate as it would when performing several t -tests. This is one definitive advantage of performing ANOVA. As we will see,

this is not the only reason for ANOVA, and it is a bit more complex than simply the type I error rate. Historically, ANOVA was responsible for providing researchers with a “toolkit” of statistical methods, of which many varieties well beyond the “classic” ANOVA have sprung. ANOVA allows for much more complex modeling that is beyond the scope of this book, and, both historically and currently, had broadened the landscape a great deal as it concerns statistical models. For example, multilevel and mixed modeling, relatively advanced subjects, are direct extensions to “basic” ANOVA models.

6.3 Understanding ANOVA Through an Example

The easiest way to introduce and understand ANOVA is by considering an example with some data. From this example the primary motivation for how ANOVA works on a theoretical level will surface. Consider the data (featured earlier in the book) on achievement scores as a function of which teacher students were randomly assigned. The data appear in Table 6.1.

Let us describe this data in some detail so that we are very familiar with it as we unpack the concepts of ANOVA. With data, especially such as these, which are rather small, you should always inspect and look at the data carefully to become familiar with it. **Always get to know your data regardless of what model you are fitting.** As concerns the current data:

- In each group are achievement scores of students who were randomly assigned to each teacher. For instance, under “Teacher 1” the first student obtained a score of 70, the second student a score of 67, and so on.
- There is a total of four teachers in the data layout.
- The mean of achievement scores for each teacher are given at the bottom of each column. The mean for teacher 1 is 71.00, the mean for teacher 2 is 72.5, and so on.
- Note that the sample means are different from one another. The first two means (71.00 and 72.50) are relatively similar, while the other two means are quite different from both each other and from the first two means. The idea of sample means being different from one another can be thought of as **between-group variability**.

Table 6.1 Achievement as a function of teacher.

Teacher			
1	2	3	4
70	69	85	95
67	68	86	94
65	70	85	89
75	76	76	94
76	77	75	93
73	75	73	91
$M = 71.00$	$M = 72.5$	$M = 80.0$	$M = 92.67$

- We note that in addition to between-group variability, there is also **within-group variability**. For example, in group 1, we note a certain degree of variability, such that 70 is different from 67, which is different from 65, and so on. We note there is within-group variability in groups 2, 3, and 4 as well.

The fundamental question that ANOVA will attempt to address of this data is the following:

Is total variation in the data better explained by between-group variation or within-group variation?

That's it! That is the underlying entire goal of the analysis of variance. The rest is in the details and complexity of each individual model. Applied to our data, ANOVA will attempt to compare **between-group variation** to **within-group variation**. But why do this? Because it is going to help us in evaluating the null hypothesis for ANOVA. The null hypothesis for ANOVA is that all population means are equal. That is, for our data, the null would be

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Note that as was true for the independent samples *t*-test, this hypothesis is about **population means**, not samples! Make sure you understand this point. Many students who claim to be familiar with ANOVA and statistical methods in general often still fail to realize that **null hypotheses are about population parameters, not sample statistics**. In our achievement data, we clearly do not have populations. We have samples. Hence, the sample means of 71.00, 72.5, etc., different as they are, would not by themselves justify rejecting the above null hypothesis. Otherwise, the null hypothesis would always be rejected for virtually all data since we always expect to see differences in any samples we can gather. The question for ANOVA is whether the differences we are observing in our sample provides any evidence that the population means are different from one another. That is the question posed by ANOVA. Again, this detail is extremely important! Make sure you understand it. **If you have population data, and hence have knowledge of the population parameters, you are no longer interested in statistical significance!** If you have the population data, a significance test would be meaningless. The only reason for conducting a significance test is because you have sample data and would like to use the sample data to get a picture of what might be going on in the population.

As an example, suppose a department of faculty members is voting on a policy. Suppose there are 20 faculty members who vote. In the actual vote, 11 are in favor, while 9 are opposed. Does it make sense to ask the question of whether the result is statistically significant? Absolutely not! Why not? **Because we have the population data. The population is the faculty.** Unless the faculty comprises a subset of a wider group of faculty members to which we wish to infer, the vote of 11 to 9 is the population result we were seeking to learn about. It just so happens that, in this case, the numbers are quite small, but they are still parameters. Hence, the concept of statistical significance has no meaning here. What may still have meaning is the idea of **measurement error**, but that is different from statistical significance. Measurement error is how precisely we are assessing or evaluating an attribute, so, in this case, it may be worth re-doing the vote to make sure measurement error is minimized and that no faculty members misunderstood the question or the item being voted on, or voted out

of fear of not conforming to the group (group conformity in a faculty meeting, who would have guessed that?). Assessing measurement error is well beyond the scope of this book and is a topic more aligned with **psychometrics** than statistical methods in general (though both areas closely overlap). If you have a significant degree of measurement error in your variables, it can severely compromise your statistical analyses.

Let's get our data into Python to have a better look (we print only a few cases):

```
data = {'ac' : [70, 67, 65, 75, 76, 73, 69, 68, 70, 76, 77, 75, 85,
86, 85, 76, 75, 73, 95, 94, 89, 94, 93, 91],
'teach' : [1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4,
4, 4, 4, 4, 4]}
```

```
df = pd.DataFrame(data)
```

```
df
```

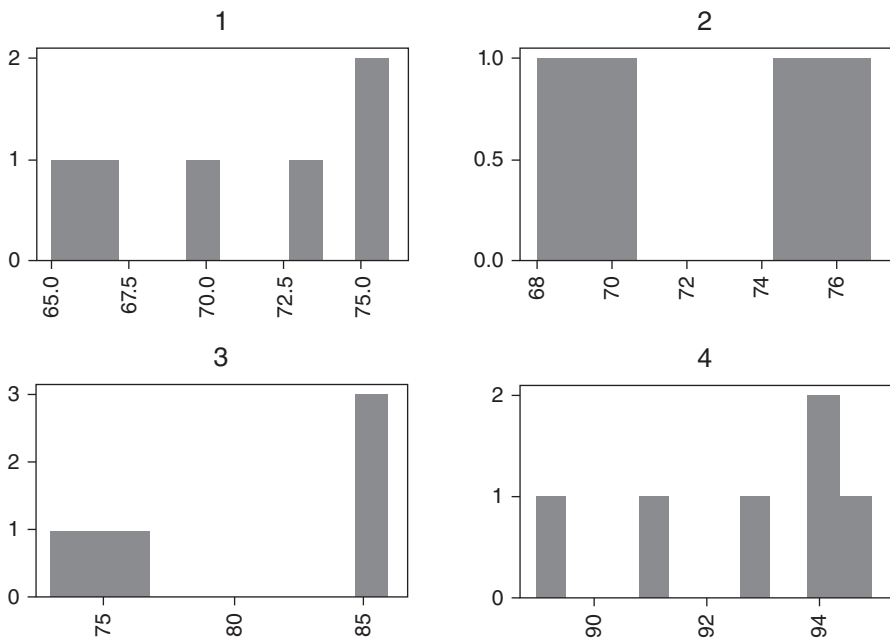
```
Out[211]:
```

	ac	teach
0	70	1
1	67	1
2	65	1
3	75	1
4	76	1
5	73	1

Visualizing our data will help us get a better idea of the distributions of achievement by each teacher:

```
df['ac'].hist(by=df['teach'])
```

```
Out[224]:
```



The histograms are a bit “ugly” because there is not a lot of data in each group, but one can still get a rough sense of how much variability there is in each of the teacher groups. Some descriptive statistics will also help. We could get the sum and mean of each variable. First, the sum is computed:

```
df.sum()
Out[228]:
ac          1897
teach        60
dtype: int64
```

Always be cautious about interpreting summary statistics, as depending on the types of variables on which you are computing, the result may be quite meaningless or misleading. For instance, we see that the sum for `teach` is equal to 60, but this means nothing to us. If you look at our `teach` variable, it is simply summing the following:

$$1 + 1 + 1 + 1 + 1 + 1 + 2 + 2 + 2 + 2 + \dots + 4$$

However, this corresponds to **levels** of `teach` as a factor and hence the total sum is a useless statistic. On the other hand, the sum of 1897 for `ac` is a bit more meaningful, since it gives us an idea of how scores across teacher groups total up. In the following, we obtain the **arithmetic mean** for both `ac` and `teach`:

```
df.mean()
Out[233]:
ac          79.041667
teach        2.500000
dtype: float64
```

The mean for `ac` is equal to 79.04. This is definitely meaningful, since it gives us the **overall** or **grand mean** of all the data. The mean for `teach` is quite useless, since it gives us the mean factor level. Again, when you ask Python to compute such things as means and standard deviations, while it is easier sometimes to not specify exact variables on which you want these computations (e.g. we could have preselected `ac` by itself had we really wanted to), you simply have to know what to interpret and what to ignore. This applies for virtually all software output. In the following we get the **standard deviations** and **medians**:

```
df.std()
Out[234]:
ac          9.648064
teach        1.142080
dtype: float64

df.median()
Out[235]:
ac          76.0
teach        2.5
dtype: float64
```


It is easier in many cases to simply request the above statistics and others using `describe()` in Python:

```
df.describe()
Out[236]:
```

	ac	teach
count	24.000000	24.000000
mean	79.041667	2.500000
std	9.648064	1.142080
min	65.000000	1.000000
25%	72.250000	1.750000
50%	76.000000	2.500000
75%	86.750000	3.250000
max	95.000000	4.000000

All of the descriptives for `ac` are useful to us. For example, we see that the minimum value in our data is equal to 65 while the maximum is equal to 95. This can be important information, since if the passing score is, say, 50, we immediately know that nobody failed. Likewise, if we know that the theoretical maximum is equal to 100, then we know nobody got the highest score possible, since the highest score is equal to 95. The 25%, 50%, and 75% values are also useful. These are the **quartiles** of the distribution. That is, 25% of scores in the data set are below 72.25, 50% are less than 76.00, and 75% are smaller than 86.75. Notice that the 50% score of 76.00 is the actual **median** of our data, and agrees with our request for the median earlier using `df.median()`. The `count` field is also of interest, as it tells us there are a total of 24 data points. This can be useful as a quick verification and check to make sure all data points are represented in our summary statistics and that no data omission errors have occurred. For the `teach` variable, again most of the information is irrelevant due to the nature of `teach` being a factor variable. However, the fact that the minimum value is 1.00 and the maximum value is 4.00 confirms for us the range on the factor level from 1 to 4. Such **confirmatory checks** can be useful to make sure Python is reading your data file correctly and that all information is being accounted for. Checking, double-checking, and re-checking those double-checks is what good data management and analysis is all about. Never hesitate to “confirm the obvious,” it may surprise you!

6.4 Evaluating Assumptions in ANOVA

Recall that as with virtually any statistical model, it is necessary to ensure that assumptions for the model are satisfied, or at least not so violated that it makes the model inappropriate and inferences misguided. Recall from the outset that assumptions for any statistical model are almost always wrong. **The job is to make sure they are within bounds.**

A first assumption of ANOVA is that of **independence of errors**, which essentially implies that errors within one group are not statistically related to other errors within the group and to errors in other groups (Kirk, 1995). This assumption is essential and can usually (though it does not necessarily guarantee) be considered satisfied by the method of data recruitment and assignment to groups. If you have a reliable means of

randomly sampling observations and assigning those observations to groups on a random non-biased basis, it is likely that the assumption will be more or less met. If you put out a flyer for participation for your study and two college students participate and share information about the experiment with one another, independence will be violated to some degree. Whether the **degree of violation** will be enough to thwart inferential tests may be unknown, but it should concern you if you have reason to doubt independence at any point in your experiment or study. Rarely in social studies is independence ever truly satisfied, but the task is to minimize it being violated as much as possible. **Residual plots** can sometimes be used to try to detect a violation of independence, but not without some difficulty. See Denis (2021) for more details.

A second assumption in ANOVA is that of **normality**. Technically speaking, the assumption refers to that of the errors of the model. However, the assumption can also be generally (though not exactly equivalently) phrased as the normality of population distributions in each level of the independent variable (IV). How to verify this assumption? Usually, visual plots of **residuals** are sufficient. For instance, plotting boxplots and histograms will usually suffice in giving you an idea of whether the assumption is more or less tenable. Some researchers (and reviewers of journal articles) like to also supplement this information with an inferential test of normality. The **Shapiro-Wilk normality test** is one such test that evaluates the null hypothesis that the data were drawn from a normal population. The technical details of the test need not concern us here and are not tremendously enlightening. What you should be aware of, however, is that the test has its own issues and can be quite sensitive to assumptions itself as well as sample size. Hence, it is generally recommended that you do not take the test “too seriously,” but instead use it as a guide only for assessing normality.

Now, for our data, we have a total of six observations per group. There is realistically no way of evaluating normality per group with only six observations; hence, conducting the Shapiro-Wilk test on each group would be pointless. However, and for demonstration here, conducting the test on the entire data set may nonetheless give us a sense of the **distribution** of the entire data. Again, we are not taking this test too seriously since it is on the entire data set and not by group, but it nonetheless can give us an idea of the global normality of the data (and allow us to demonstrate the Shapiro-Wilk test). To get the test, we import **scipy** and use the `stats.shapiro()` function:

```
import scipy as sp

ac = df['ac']
sp.stats.shapiro(ac)
Out[241]: (0.90565425157547, 0.028417399153113365)
```

The value of the test statistic is equal to 0.90, with a *p*-value equal to 0.028. Recall that the null hypothesis is one of normality. Hence, since the *p*-value is quite small, it indicates that we can reject the assumption of normality and infer non-normality. That is, we have evidence to suggest that the sample of data was not drawn from a normal population. Again, with larger data sets the test will be more informative and you should conduct the test **per group or cell** to get an idea of normality, not on the entire data set. And to reiterate, use the test (if you use it at all) as a **guide** only, not a final verdict. We have more to say on assumptions for parametric models later in the chapter, where we discuss this issue at some length and how you should ideally approach it from both a statistical and scientific perspective.

A third assumption that needs to be satisfied when conducting ANOVA is that of **homogeneity of variance**. That is, ANOVA assumes that variances in each population are equal to one another. The null hypothesis is thus

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$$

where σ_1^2 is the variance for population 1, σ_2^2 is the variance for population 2, and so on. Since we have a total of four populations, we have a total of four variances to verify. Now, a moment's contemplation will reveal that, just like with virtually any null hypothesis significance test, **the null is dead in the water even before conducting the test!** That is, surely the population variances will not be equal, which we know long before conducting the test! They will be different to at least some degree, to some decimal place, even if they are quite similar to begin with. So why conduct the test at all? Why evaluate an assumption we already know to be false? For the same reason why we do so with means. We know the null is, in theory, a priori false, yet we conduct the test anyhow as a test against an “ideal” model to learn whether the data differ sufficiently to reject the hypothetical model. To evaluate equality of variances, we can conduct several tests, a popular one being that of **Levene's test for the equality of variances**:

```
import scipy as sp
teach = df['teach']
sp.stats.levene(ac, teach)

Out[247]: LeveneResult(statistic=26.34364367460562,
pvalue=5.625609083376561e-06)
```

The value of the test statistic is equal to 26.34. The p -value for the test is very small, suggesting that we can reject the null hypothesis of equal variances. What the p -value is saying here is that the probability of the data we are witnessing in terms of variances is very small under the hypothetical situation of equal population variances. Hence, since the probability of this data is so small, we deem it unlikely that it could have arisen from a model in which the variances are equal. We reject the null hypothesis and conclude that the variances are unequal. However, since ANOVA is quite robust to the equality of the variance assumption and for the sake of demonstration, we choose to push forward with the ANOVA anyhow and interpret it. In practical cases, if you do have wildly different variances, it behooves you to look further at your data in some detail. **Get curious about your data.** There may be a good reason why your variances are unequal and it may be of scientific interest, in addition to statistical, to learn why. For example, if it were found that improvement in COVID-19 symptoms were much more variable for individuals receiving a heavy dose of a drug than a low dose, this in itself may be an interesting result, and may be a reason to follow up on the finding. Inequality of variances may be itself an interesting phenomenon. **Do not simply seek to appease a statistical model with the assumption. A violation of variances could indicate something of scientific interest. Be more curious about your data than simply satisfying assumptions!**

6.5 ANOVA in Python

We now push forward with the ANOVA in Python, exploring a couple ways in which we can obtain output. We also interpret the output in great detail to learn exactly what each piece is telling us. Before conducting the actual ANOVA, it can be useful as a data check to compute `info()` to verify some basic logistics:

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24 entries, 0 to 23
Data columns (total 2 columns):
ac          24 non-null int64
teach       24 non-null int64
dtypes: int64(2)
memory usage: 464.0 bytes
```

We now run the ANOVA model using the `ols()` function:

```
import statsmodels.api as sm
from statsmodels.formula.api import ols
model = ols('ac ~ C(teach)', data=df).fit()
```

Notice that we named our object `model` and specified the formula as `ac ~ C(teach)`, where `C(teach)` is telling Python to treat it as a categorical variable. To get the ANOVA summary table, we compute the following, where we also request **Type 2 sums of squares** (you will obtain the same in this case had you chosen `typ=3`):

```
table = sm.stats.anova_lm(model, typ=2)
print(table)
```

	sum_sq	df	F	PR(>F)
C(teach)	1764.125000	3.0	31.209642	9.677198e-08
Residual	376.833333	20.0	NaN	NaN

Let us unpack each component of this summary table:

- The ANOVA has separated components of **sums of squares** into two parts, that due to `teach` and that left over in `residual`. The sums of squares for `teach` are equal to 1764.125, while the residual sums of squares are equal to 376.833. The total sums of squares is therefore equal to $1764.125 + 376.833 = 2140.958$.
- Each sum of squares has associated with it **degrees of freedom**. The degrees of freedom for `teach` are equal to one minus the number of groups. Since there are four groups, the degrees of freedom for `teach` are equal to 3.0. For within, we lose a single degree of freedom per group across all groups. Since there are four groups, this yields a total of 20 degrees of freedom.
- The **mean squares** for ANOVA are not reported in the table, though they are implicit, since they are used to compute the resulting *F*-ratio. The mean squares, or “MS” for short, are computed as SS/df , which in our case for `teach` are equal to $1764.125/3 = 588.04$. The mean squares for `residual` are equal to $376.833/20 = 18.84$.
- The ***F*-statistic** for the ANOVA recall is computed as a ratio of MS between to MS within, so, for our data, the computation is $588.04/18.84 = 31.212$, which is evaluated as coming from an *F* distribution on 3 and 20 degrees of freedom.

- The ***p*-value** resulting from the significance test on *F* is equal to 9.677198e-08, which when translated out of scientific notation, equals 0.00000009677198. Hence, we can see it is much less than a conventional level set such as 0.05, and hence we deem the *F* statistic statistically significant. That is, we have evidence to reject the null hypothesis and conclude that somewhere among the means there is a mean population difference.
- Simply because we have rejected the null, however, does not inform us of where exactly the mean differences may lie. The *F*-test is an omnibus test such that it gives us the overall test, but it does not tell us specifically which means are different from one another. For that, we will require a **post-hoc test**.

6.6 Effect Size for Teacher

Having conducted the ANOVA, we would now like to obtain, in addition to the observed *p*-value, a measure of **effect size**. Remember, *p*-values do not tell the full story and in many cases can be quite useless as an indicator of scientific evidence. Much more meaningful is the size of effect. So how should we compute the effect size in this context? Though a variety of effect sizes can be computed, by far the most common is to simply compute an **eta-squared** statistic, often denoted by the symbol, η^2 . Eta-squared is given by

$$\text{SS treatment} / \text{SS total}$$

Since we know that the total variation of all the data is contained in SS total, it makes good sense to take this ratio as the proportion of variation or variance attributable to treatment. In our case, the treatment effect is the teacher effect. Hence, for our data, we compute η^2 to be $\text{SS teach} / (\text{SS teach} + \text{SS residual}) = 1,764.1 / (1,764.1 + 376.8) = 0.82$. We interpret the number 0.82 to mean that approximately 82% of the variance in achievement scores can be attributed to teacher differences. Is that effect large? In a sense, it depends on the research area. Effect sizes should be interpreted relative to the research area from which they are originating. However, if I told you that depending on which teacher one had in school, it could explain upward of 82% of the variance in their scores, you would probably take a great interest in who teaches your son or daughter! Eta-squared it turns out is a bit biased upward, meaning that the size of the statistic is somewhat of an overestimate of the actual effect size in the population. A less biased estimate of effect size is **omega-squared**, though not discussed here (see Kirk, 2012, for details). In terms of a ballpark estimate, eta-squared usually does fine for describing an effect, and will give you a quick sense of what “happened” (if anything) in the study or experiment.

6.7 Post-Hoc Tests Following the ANOVA *F*-Test

The ANOVA conducted earlier yielded evidence of population mean differences between levels on the IV. However, the *F*-test does not address specifically which means are different from one another, only that somewhere among the means there is a difference. If a researcher has an idea in advance of conducting the analysis where mean differences are likely to lie, then that researcher has the option of preparing and performing a **priori contrasts**. For example, if the researcher hypothesized that means for teachers 1 and 4 would be different (perhaps they have completely

contrasting teaching styles), then planning to do a t -test to compare these groups can be set out long before the data are collected and obtained. Comparisons or contrasts such as this are known as “a priori” because they are **planned before looking at the data**. Even if the data are archival, the key point is that you have not yet “looked at them.” Thus, when you conduct one or two contrasts, you can be relatively assured that your error rate is not skyrocketing. However, if you look at the means first, then it is most likely that you are performing well more than simply that comparison. Let us unpack this concept a bit further, as it is an important point, as the rationale behind the post-hoc concept can seem at first quite mysterious. Suppose you had no prior theory as to which means might be different from other means for the given study. Having collected the data, you then list the means featured in the data table discussed earlier:

71.00 72.50 80.00 92.67

Now, having looked at the means, and if asked to make one pairwise comparison among them, which one would you make? You would likely choose to compare the first mean to that of the fourth mean. That is, you would compare 71.00 to 92.67. Why? Simply because that is without a doubt your best chance at obtaining a statistically significant pairwise comparison. Since these means are the most disparate, they stand pretty much the best chance at rejecting the null for a pairwise comparison. However, even if you choose to conduct this comparison, how many comparisons did you really make? You did not make just one. You made several more. When you looked at the means, you most likely compared the first mean to the second, the second to the third, the third to the fourth, and so on. That is, **it is likely that you made many more mean comparisons than the one you chose to perform**. Now, you did not actually conduct these comparisons formally, but you did so **mentally**, and it stands that any good test of means should in some sense incorporate these numerous comparisons into its computation. This is exactly what post-hoc tests attempt to do. A post-hoc test assumes from the outset that you are performing more than one or two comparisons. Post-hocs generally assume that you are performing, at least visually, virtually all pairwise comparisons possible on the data.

What is wrong with performing this many comparisons? As discussed earlier with respect to the t -test, for each comparison made, there is a type I error rate to be concerned about. And, across the board, the type I error rate will begin to rise sharply. Hence, when you compare means 1 to 4, you are not evaluating this difference at a familywise error rate of 0.05. Instead, you are conducting this comparison at a much higher familywise error rate. Often this error rate is unknown, because nobody knows how many comparisons you actually did do in choosing the comparison of interest. Perhaps you simply did pairwise comparisons. However, you could have possibly also compared means 1 and 2 to mean 3. Hence, a good post-hoc test will attempt to mitigate the overall and inflating type I error rate across the numerous comparisons you may in reality be making. Or, sometimes you may wish to simply “snoop” the data to learn where the differences actually are, and have no prior theoretical inclination as to where these differences may be. In both cases, post-hoc tests are appropriate.

6.8 A Myriad of Post-Hoc Tests

There are a huge number of post-hoc tests a researcher can choose from when shopping for a test. What differentiates them, in general, is how they go about controlling the type I error rate. Some tests attempt to guard against inflation for all pairwise comparisons simultaneously, while other tests do so in an incremental stepwise fashion. Choose one test and you may find statistical evidence for a difference. Choose another and you may possibly not. This is the nature of post-hoc tests.

Now, you may at first ask yourself: “If there is a difference in means, shouldn’t any and all post-hoc tests ‘find it?’ For example, in our achievement data, shouldn’t the mean difference between teacher 1 and, say, teacher 3, of 71.0 vs. 80.0, be ‘detectable’ by any test? After all, it is clearly a difference, so why should it matter what post-hoc test is used to detect it?” If you are asking this question, it suggests you do not quite yet understand how statistical inference works. Recall these differences are on **samples**, not populations. When we observe a difference of 71.0 vs. 80.0, though this is clearly an arithmetic difference in the sample, we do not yet know whether a difference exists in the population from which these data were drawn. This is why we need to do the significance test. We are evaluating whether μ_1 is different from μ_3 , which is the question we are asking, not whether there are differences in the sample. Remember, this is how statistical inference operates in general. Unless we have the population parameters at our disposal, we are using the sample information to make a “guess” at what might be going on in the population. Hence, it makes perfect sense that different post-hoc tests will potentially generate different conclusions on the null $H_0: \mu_1 = \mu_3$, for instance. It is how those different tests mitigate or regulate type I error rates and statistical power that is the issue. **A difference in the sample does not necessarily suggest a true difference in the population. This is the precise reason why statistical inference is necessary.**

This is about as far as we explore the logic of post-hoc tests in this book. There are many excellent treatments of post-hoc tests, including Howell (2002), which is highly recommended for a very good overview. For our purposes, we cut to the chase and will stick with recommending one of the most common and reliable tests, that of the **Tukey Honestly Significant Difference** test, or **TukeyHSD**, for short. It does a good job at controlling inflating type I error rates, without sacrificing too much power. If you would like an even more stringent and conservative test (meaning that it will guard even more against type I error, but at the expense of lower power), then the **Scheffé test** is recommended. The Scheffé is often a choice of hardliner scientists who only want to reject the null if there is relatively strong evidence that the null is false. However, as you can imagine, the test often lacks power and hence if you are trying to learn if a treatment for COVID-19 is effective, for instance, the Scheffé test might miss it. We should also mention the Bonferroni adjustment, discussed earlier. Recall that this is when family-wise alpha is divided by the number of pairwise comparisons you are making. This test can sometimes be used as a post-hoc as well. For example, if you wish to make three comparisons at a family-wise error rate of 0.05, then you would be testing each comparison at $0.05/3 = 0.016$. As you can imagine, as the number of

comparisons increases, power begins to suffer greatly. For example, for 10 comparisons, we would be testing each comparison at $0.05/10 = 0.005$, which does not afford us very much power to reject the null for any comparison. Hence, use the Bonferroni adjustment judiciously as a post-hoc test. This actually goes for all post-hoc tests; try out a few of your favorites on your data and see what they say. If they report conflicting results, it may be that the differences you would like to think are present are not quite so strong.

We can easily conduct the Tukey test on our achievement data by the following. We will name the object `post` and request `tukeyhsd()`:

```
from statsmodels.stats.multicomp import (pairwise_
tukeyhsd, MultiComparison)
post = pairwise_tukeyhsd(df['ac'], df['teach'])
print(post)
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj   lower   upper   reject
-----
      1      2       1.5    0.9 -5.5147   8.5147   False
      1      3       9.0 0.0091  1.9853  16.0147    True
      1      4    21.6667  0.001 14.6519  28.6814    True
      2      3       7.5 0.0335  0.4853  14.5147    True
      2      4    20.1667  0.001 13.1519  27.1814    True
      3      4    12.6667  0.001  5.6519  19.6814    True
-----
```

Let us unpack what this output is telling us:

- Python tells us it is conducting a multiple comparison of means, where `FWER = 0.05` is the familywise error rate.
- In the actual output are the pairwise comparisons. We see that for group 1 vs. group 2, the mean difference is equal to 1.5, with an associated *p*-value of 0.9. Since the *p*-value is so high, the decision is to not reject the null hypothesis. The output also includes confidence limits on the difference, which for this particular comparison ranges from -5.5147 to 8.5147.
- For group 1 vs. group 3, the mean difference is equal to 9.0, with an associated *p*-value of 0.0091. Since the *p*-value is so small, definitely less than a cut-off value of 0.05, we reject the null hypothesis. This is what is indicated by Python under `reject = True`.
- The remaining group differences are given in the output and are interpreted analogously. The largest difference is between that of group 1 vs. group 4, a mean difference equal to 21.6667 with an associated *p*-value of 0.001. Hence, we can see that for all pairwise differences, the only sample difference not statistically significant according to the Tukey test is that between groups 1 and 2. All the other pairwise differences are statistically significant, suggesting that in the population from which these data were drawn, there are mean differences.

6.9 Factorial ANOVA

Our survey of the analysis of variance thus far has focused on the one-way ANOVA. That is, in our earlier analysis, we only had a single dependent variable (DV) and IV. The goal was to learn of any mean differences on that single IV. In many cases, however, a researcher has more than a single IV and may wish to model them simultaneously. For the example we will use, suppose that instead of being interested in whether there are mean achievement differences across teachers, the researcher was also interested in learning whether there might be mean differences across teachers and textbooks used (Table 6.2). Suppose that in addition to students being randomly assigned to a teacher, they were also randomly assigned to different textbooks, as shown in the following table (notice that “Textbook” is now the new variable in the rows of the layout):

Table 6.2 Achievement as a function of teacher and textbook.

Textbook	Teacher			
	1	2	3	4
1	70	69	85	95
1	67	68	86	94
1	65	70	85	89
2	75	76	76	94
2	76	77	75	93
2	73	75	73	91

Thus, the primary question asked by the researcher is now the following:

Is achievement a function of teacher and textbook considered simultaneously?

Now, the wording here matters. The “simultaneously” matters a great deal. Why? Let us demonstrate. Recall in our ANOVA we found evidence for teacher differences. Would our model have detected this same effect had we also included textbook as a factor? Let us find out. In the following, we model both teacher and textbook in the same model:

```
data = {'ac' : [70, 67, 65, 75, 76, 73, 69, 68, 70, 76, 77, 75, 85,
86, 85, 76, 75, 73, 95, 94, 89, 94, 93, 91],
'teach' : [1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4,
4, 4, 4, 4],
'text' : [1, 1, 1, 2, 2, 2, 1, 1, 1, 2, 2, 2, 1, 1, 1, 2, 2, 2, 1,
1, 1, 2, 2]}
df_2 = pd.DataFrame(data)
df_2
```

```

Out[288]:
      ac  teach  text
0     70      1     1
1     67      1     1
2     65      1     1
3     75      1     2
4     76      1     2
5     73      1     2
. . . . .

import statsmodels as sm
import statsmodels.api as sm
model_2 = ols('ac ~ C(teach) + C(text)', data = df_2).fit()
table_2 = sm.stats.anova_lm(model_2, typ = 2)
print(table_2)

              sum_sq      df      F      PR(>F)
C(teach)    1764.125000     3.0   30.051216  2.029775e-07
C(text)         5.041667     1.0    0.257649  6.175816e-01
Residual    371.791667    19.0         NaN         NaN

```

Notice carefully that now that we have added the second factor `text`, **the output line for teach is not the same as it was in the one-way model!** The sums of squares for `teach` are the same, the degrees of freedom are the same, but the resulting F -statistic is different. In the one-way model, recall our F and corresponding p -value came out to be 31.21 and 9.677198e-08, respectively. When we include the `text` factor, the F -statistic now comes out to be 30.05 with the p -value 2.029775e-07. Because the p -value is now larger in the two-way than in the one-way model, we have lost some power. But why? More specifically, **why are we seeing one effect in one model and another effect in the other?** Though both are clearly recommending a rejection of the null hypothesis, it is equally clear that the p -values are different.

The reason this is occurring is attributable not only to how data analysis works in ANOVA, but how statistical analysis works in virtually all statistical models. The key phrase is “variance partitioning.” In any model, we are partitioning sources of variance into components. When we performed the one-way model, we partitioned variance into that attributable to `teach` and residual. When we do the two-way, we are partitioning variance into that attributable to `teach`, `text`, and residual. Though SS total remains the same in each model, **the MS values will change because of the new partitioning.** Hence, the p -value for `teach` is not the same as it was in the original model. Let us summarize the primary outcome of this discussion, and that is this: **As a researcher and scientist, you need to be aware of this principle. When you test and evaluate effects in a statistical model, you are always evaluating them in the presence or context of other effects also included in the model. What might be statistically significant in one model may not be in another.** Of course, the scientific ramifications and implications of this are huge; in one model the treatment has an effect on COVID-19, but in another model, less so! It is vital that you look deeper into how a given model was constructed before interpreting effects. This is also why models in the social sciences can sometimes be

explained as simply being **statistical artifacts**, an exercise in variance partitioning, with no clear case to be made that any variable has an effect on any other. As you learn to appreciate how statistical models function, you will no longer look at research findings in the same way, and will look for evidence that the material under discussion (the “stuff” on which the statistics are being computed) has in fact changed due to the treatment, rather than merely be satisfied with a change in statistical coefficients.

6.10 Statistical Interactions

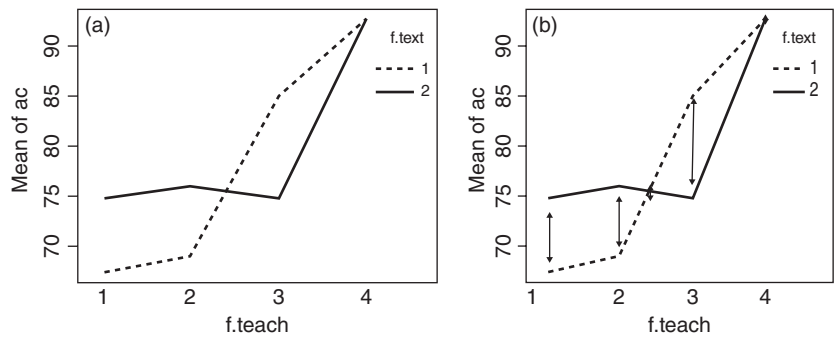
Having above added textbook to the model, the real reason for conducting a factorial ANOVA over two one-way ANOVAs is not simply to add an additional factor. The real reason is usually to seek out **statistical interactions**. What is an interaction? An interaction between factors is said to exist when **the effect of one IV on the response is contingent upon the level of the second factor**. As a COVID-19 example, suppose a treatment for the disease was found to be effective, but this effectiveness was different for males than it was for females. For instance, suppose it worked well for females, but not males. Then we would conclude that the treatment works, but not quite as well for males as for females. This is the nature of an interaction effect. Hence, in our achievement as a function of teacher and textbook layout featured earlier, we are no longer merely interested in the effect of teacher and of textbook independently. We are instead interested in the **cell means** primarily as they give us a sense of whether an interaction may be occurring in our data. For example, in the following (Table 6.3) we specifically emphasize the cell means (each cell enclosed in a rectangle):

Table 6.3 Achievement as a function of teacher and textbook.

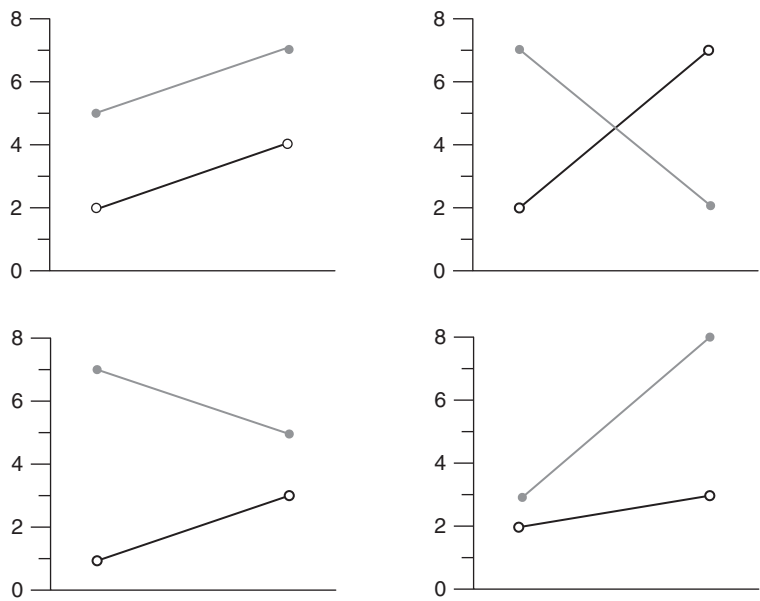
Textbook	Teacher			
	1	2	3	4
1	70	69	85	95
1	67	68	86	94
1	65	70	85	89
2	75	76	76	94
2	76	77	75	93
2	73	75	73	91

Let’s look at the above cell means very carefully. Notice that, in general, as we move from teacher 1 through 4, there is a general inclination for means to increase. However, this is primarily only true for textbook 1. Look at textbook 2. For textbook 2, though there is a slight increase from teacher 1 to 2, there is then a decrease to

teacher 3, then up again to teacher 4. Hence, **the effect of teachers on achievement is not consistent across textbooks**. This suggests the existence of an interaction effect. Interaction plots are by far the easiest way to detect the presence or absence of interaction effects, as we can plainly see in the following plot of the achievement data.



Closer inspection of these plots will reveal that the presence or absence of an interaction is indicated by whether or not the lines are **parallel**. If the lines are perfectly parallel, then there is no interaction in the sample, since the same “story” is being told regardless of which textbook is being considered. As an example of a generic plot in which there is no interaction, consider the following graph, top left cell.



Note that the lines in the top left cell are perfectly parallel. All of the other plots suggest the presence of an **interaction in the sample**, most notably the complete crossing of lines in the top right cell. This brings us to an important question: **Can an interaction in the sample ever not exist?** We consider this possibility next.

6.11 Interactions in the Sample Are a Virtual Guarantee: Interactions in the Population Are Not

There is an important distinction between an interaction in the sample versus one in the population. An interaction in the sample is virtually a guarantee for any data set you may consider. How so? Because just as there is a virtual guarantee of sample mean differences in a t -test or ANOVA to some decimal place, **non-parallel lines in an interaction plot will virtually always exist**. Think about why this is so. Do you really think if we plotted an interaction graph in a sample that the lines could ever be perfectly parallel? Hardly ever, unless we were dealing with a very contrived situation or data set, or rounding to whole numbers or something. Otherwise, lines will always be **non-parallel**, at least to some degree in the sample. Of course, our interest, as always, is not in the sample, but rather the population. Whether the sample data suggest the presence of an interaction in the population is what we are truly interested in. For that, we require the significance test on the interaction term. However, in the sample, you will virtually always see an interaction.

The author has observed many times researchers presenting interaction graphs in which there is clearly an interaction in the sample, yet standing next to the graph claiming there is no interaction! Of course, they were right to say they had no evidence of an interaction in the population. However, in most cases, there is clearly an interaction in the sample. Unpacking the distinction here to your audience is of vital importance, but never stand next to an interaction plot in which the lines are clearly non-parallel and say there is no interaction if there clearly is. As with all significance tests, as sample size increases, the p -value for the interaction effect will typically get smaller, and hence eventually, if the effect remains more or less constant, you will reject the null hypothesis of no interaction.

6.12 Modeling the Interaction Term

Earlier, we fit the ANOVA model that featured both main effects. However, we did not specifically model the interaction term. We can model the interaction term as follows by including the term `C(teach)*C(text)`:

```
import statsmodels.api as sm
from statsmodels.formula.api import ols
model_3 = ols('ac ~ C(teach) + C(text) + C(teach)*C(text)', data =
df_2).fit()
table_3 = sm.stats.anova_lm(model_3, typ = 2)
print(table_3)
```

	sum_sq	df	F	PR(>F)
C(teach)	1764.125000	3.0	180.935897	1.488709e-12
C(text)	5.041667	1.0	1.551282	2.308781e-01
C(teach):C(text)	319.791667	3.0	32.799145	4.574415e-07
Residual	52.000000	16.0	NaN	NaN

We unpack each component of the output:

- The two-way factorial has separated the sources of variability into an effect for `teach`, `text`, the interaction of `teach` by `text`, and `residual`.
- We begin by interpreting the interaction term and see that it is statistically significant, yielding a very small p -value ($p = 4.574415e-07$). This suggests that any effect of `teach` on achievement is not consistent across `text`, and that any effect of `text` is not consistent across `teach`. Ideally, **interaction terms should be interpreted first over and above main effect terms**. Why? Because they tell more of the “story” of the data. While it is technically not completely incorrect to interpret main effects in the context of an interaction, in light of an interaction, main effect terms will be misleading as they do not tell you the full story (e.g. if there is a treatment effect and it is only for older people and not young, the main effect term is still correct to say there is a treatment effect overall, but you (and your readers) are being misguided by interpreting the main effect only).
- Both main effects of `teach` and `text` are both statistically significant, yielding p -values of $1.488709e-12$ and $2.308781e-01$, respectively.
- The residual sums of squares for the model are equal to 52 on 16 degrees of freedom. No F -statistic or p -value is computed for the residual.

It is pedagogical to compare the two-way ANOVA fit earlier, where no interaction was modeled, to the two-way ANOVA we just fit with the interaction term. Notice that the total sum of squares is the same in both cases:

Main Effects Only: $1,764.125000 + 5.041667 + 371.791667 = 2,140.958334$

Interaction Model: $1,764.125000 + 5.041667 + 319.791667 + 52.000000 = 2,140.958334$

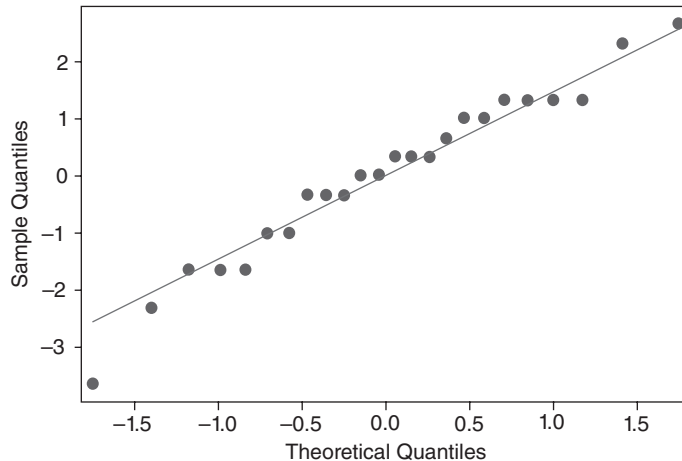
The only difference in the two models is that SS residual in the first model (i.e. “main effects only”) was partitioned further into SS interaction in the second model. That is, notice that the SS residual of 371.791667 in the main effects model is equal to the sum of residual + interaction in the interactive model (i.e. $319.791667 + 52.000000 = 371.791667$). To fit the interaction, the ANOVA “stole” from the residual term, that is, it took from the portion of variance explained, and attributed some of that to interaction. This is how modeling in general works, partitioning sources of variance into components. But then why have the p -values changed? The p -values are different because the mean squares values have changed as a result of including the interaction term. **The model you test determines the effects that you will or will not see.** This is well worth reminding yourself of repeatedly as you interpret research. It is a core principle of statistical modeling. Hence, when you read a research claim attesting that a given variable has an effect on another, your concern is the model in which the given effect was found, since all models are **context-dependent** in this sense. This principle is true of virtually all statistical models, not only ANOVA. It is likely to be the most important principle you can take away, as a scientist, from this book! We will revisit this issue further when we discuss regression models a bit later in the book.

6.13 Plotting Residuals

Recall that we have mentioned earlier the importance of satisfying model assumptions when fitting any statistical model, and that though we can examine assumptions to

some degree by plotting raw data, it is the **residuals** we are most interested in observing for any odd or abnormal behavior. We can easily plot the residuals from our factorial model above using the following **quantile plot**. We fit the plot using **statsmodels'** `qqplot()` function:

```
import statsmodels.api as sm
res = model_3.resid
fig = sm.qqplot(res, line='s')
```



This residual plot plots **sample quantiles** against **theoretical quantiles** that would be expected under a **normal density**. A perfectly normal distribution would be represented by all dots falling along the line. Since dots deviate to some degree, we know the distribution is not perfectly normal. However, the deviation is so slight that this residual analysis is not a serious threat to model assumptions (though the residual in the lower left may be worth investigating a bit further).

6.14 Randomized Block Designs and Repeated Measures

We close our discussion of ANOVA with a brief look at **randomized block designs**, of which **repeated measures ANOVA** is a special case. Our treatment here is very cursory, as entire books are written on repeated measures models and their intricacies in modeling longitudinal data. For a good overview, see Denis (2021), which also features a fairly elaborate discussion of **random effects** and **mixed models**, of which the randomized block and repeated measures models can be considered special cases.

In the ANOVA models we have surveyed up to now, it has been assumed that individuals serving under one condition or level of the IV were different from individuals serving under other conditions. Hence, it was reasonable to expect no correlation among groups on levels of the IV. We randomly assigned individuals to each condition, hoping that any “nuisance” factors would more or less balance out across conditions. If there were a lot of nuisance factors at play, then these would pile up in the error term

of our analysis, **MS error**. If you are not randomly assigning to conditions, your design is not actually experimental. Rather, it is more akin to **quasi-experimental** and the conclusions you can draw from it are much more limited. Many statistical models, sophisticated as they are, if used without good experimental design, end up not revealing nearly as much from the data as the researcher might have hoped. Implementing a good design is half the battle to good science. The statistical model is not nearly as important. **Design first. Statistics second.**

There are times where hoping nuisance factors “balance out” in this way is not the most efficient strategy, and we can do better by purposely controlling for one or more nuisance factors by pulling such variability out of our data. To do this, we can **match participants** on one or more nuisance factors. These **matched participants** are known as **blocks**. Recall the example of a matched pairs design layout featured earlier in the book (reproduced in Table 6.4):

Table 6.4 Matched pairs design.

	Treatment 1	Treatment 2
Block 1	10	8
Block 2	15	12
Block 3	20	14
Block 4	22	15
Block 5	25	24

In the table, different participants serve under each treatment, but the **blocking factor** makes the individuals within each block more similar than had they simply been randomly assigned without the blocking factor. That is, data point 10 in block 1 is assumed to be related to data point 8 in block 1. Likewise, data point 15 is assumed to be related to data point 12 in block 2, and so on. What is the advantage of a blocking design? Quite simply, it is to extract variation due to block and remove it from the overall error term for the ANOVA. That is, the blocking factor is removed from the error term and thereby provides a more sensitive (powerful) test of the null hypothesis for the main effect. Blocking designs are especially popular in areas where rigorous experiments are regularly conducted. We do not treat block designs any further in this book. Our purpose in mentioning them at all is that they serve as the stepping stone to a model we do consider next, that of the repeated-measures model. For details on randomized blocks, see Kirk (2012).

Repeated measures is nothing more than a randomized block design in which each block represents a subject, and measurements are taken for each subject under each condition. Hence the reason why understanding block designs is mandatory to understanding the repeated-measures model. Referring back to our randomized block layout, it becomes a repeated-measures design when we replace block 1 with subject 1, block 2 with subject 2, and so on. Recall our rat data featured earlier (see Table 6.5)

Table 6.5 Learning as a function of trial (hypothetical data).

Rat	Trial			Rat Means
	1	2	3	
1	10.0	8.2	5.3	7.83
2	12.1	11.2	9.1	10.80
3	9.2	8.1	4.6	7.30
4	11.6	10.5	8.1	10.07
5	8.3	7.6	5.5	7.13
6	10.5	9.5	8.1	9.37
Trial means	$M = 10.28$	$M = 9.18$	$M = 6.78$	

Let us run the above repeated measures in Python. First, we enter the data:

```
rat = pd.DataFrame({'rat': np.repeat([1, 2, 3, 4, 5, 6], 3),
                    'trial': np.tile([1, 2, 3], 6),
                    'time': [10.0, 8.2, 5.3,
                             12.1, 11.2, 9.1,
                             9.2, 8.1, 4.6,
                             11.6, 10.5, 8.1,
                             8.3, 7.6, 5.5,
                             10.5, 9.5, 8.1]})
```

```
rat
```

```
Out[410]:
```

```
   rat  trial  time
0    1      1  10.0
1    1      2   8.2
2    1      3   5.3
3    2      1  12.1
4    2      2  11.2
5    2      3   9.1
6    3      1   9.2
7    3      2   8.1
8    3      3   4.6
9    4      1  11.6
10   4      2  10.5
11   4      3   8.1
12   5      1   8.3
13   5      2   7.6
14   5      3   5.5
15   6      1  10.5
16   6      2   9.5
17   6      3   8.1
```

There are different options in Python for running a repeated measures. We chose to use `statsmodels.stats.anova` and `AnovaRM`:

```
from statsmodels.stats.anova import AnovaRM
print(AnovaRM(data = rat, depvar = 'time', subject = 'rat', within
= ['trial']).fit())
```

```

Anova
=====
      F Value Num DF  Den DF  Pr > F
-----
trial 72.6196  2.0000 10.0000 0.0000
=====
```

The F -stat associated with the ANOVA is 72.6196, evaluated on 2 and 10 degrees of freedom. Clearly, the obtained p -value is very small (e.g. $p < 0.001$, though never **exactly** equal to 0), and hence we can reject the null hypothesis that mean time is equal across trials. We have inferential evidence that the mean on the response variable is not the same across trials.

6.15 Nonparametric Alternatives

In this final section of the chapter, we briefly survey a couple of the more common **nonparametric** (nonpar) tests that may be of use to the research scientist, and demonstrate how such tests can be run in Python. What is a nonpar test? To understand what a nonpar test is, we first need to understand the nature of a parametric test. **A parametric test is one for which before the test is performed, certain assumptions about the population from which the data were collected are assumed to be met and satisfied.** For example, in the ANOVA model just surveyed, we have had to assume such things as **normal populations, equal variances, independence of observations**. These are elements of the test that, ideally, should be true before the test is run, otherwise the results of the test will not be as valid as they could be if the assumptions were met. This is true not only of ANOVA, but also regression and a lot of other simpler statistical tests such as t -tests and so on. In the computation of F -tests, p -values, confidence intervals, etc., it was implicit that these assumptions were more or less satisfied. If they were not, or they were violated to some extent, it could compromise the validity of the test, and hence potentially thwart ensuing substantive conclusions.

Nonparametric tests are tests developed that do not make as many or as **deep assumptions** about populations as do parametric ones. Though some authors define these tests as “distribution-free” tests, other authors categorize them more as “distribution-free-er” tests (e.g. see Kirk, 2007), since they still make some assumptions, but less so than traditional parametric tests. So why is there the necessity for nonpar tests? Simply put, it is because, in most cases, assumptions about populations are either **difficult or impossible to satisfy**, and hence the assumption of an underlying probability model is often unrealistic. It takes only a moment’s reflection to realize that satisfying parametric assumptions in a perfect sense exists only in the ideal. That is, in

99.9% of cases, assumptions for virtually all statistical tests will be violated to some degree, and hence at first glance it would seem like nonpar tests should be used in all cases for all data. However, there is no need to be this absolute. Even under conditions of mild to moderate departure of assumptions, parametric tests are usually still advised over their nonpar counterparts, especially if sample size is at least minimally sufficient. However, in cases where violations are more severe, or sample size is quite small, a nonpar test may be more suitable.

When are distributional assumptions virtually guaranteed to be unsatisfied, or at minimum, unknowable? While this can be true for any data, it is especially the case for data on **small samples**. For example, with a sample size of say, six observations, if we cannot assume a priori that the sample was drawn from a normal population, then we have no way of knowing whether normality is satisfied. Plot the sample data you say? Investigating normality assumptions only works in cases where you have sufficient data to make the exploration worthwhile, such that you will actually be able to learn something about the population under investigation in the plot. In the case of six observations (again, to use our example), plotting the data will not tell you much about your distribution other than whether you have at least **some variability** in your small data set. However, if you cannot assume normality before plotting your data, the plot will not really help you either way. Hence, one common use of nonpar tests, historically, has been to use them when sample data are quite small in number and the researcher seeks a relatively quick and easy way to evaluate their data without having to invoke parametric assumptions. Many of the simpler nonpar tests are easy to conduct, simple in structure, and can readily inform the researcher of the presence or absence of a potential scientific effect. Historically, nonpar tests were also quite common before computers, since they allowed a scientist easy computation and to immediately get a glimpse of whether or not their experimental treatment had an effect. It allowed them for a simple way to exclude chance from their findings.

In many cases, one potential drawback (though this depends on what you are seeking to actually learn from your data) to using a nonpar test is that they often work on **ranking data** rather than treating the data as originally continuous. This is not an inconsequential side effect of using a nonpar test, and researchers using them need to be fully aware of how the test is treating their data. It can have fairly significant consequences. For example, recording IQs of 120, 100, and 80 is quite different from ranking them first, second, and third. After all, IQ values of 120, 119, and 118 also hold the same ranking as those of 120, 100 and 80. When a test ranks data and ignores the pseudo-continuous scale on which they were originally conceived, make no mistake, it can sometimes have pretty important consequences for how the results turn out. That is, the response to the question “**Does the treatment have an effect?**” may be different depending on whether one conducts a parametric or a nonpar test. Such differences in substantive and scientific conclusions should definitely be of concern. This is why it is essential to understand the difference between these types of tests so you can make an **informed choice** and interpretation of what the data are actually telling you. Otherwise, you are making conclusions about your data without an appreciation of what the results are based on, akin to making a diagnosis of COVID-19 without understanding how the test works. Your diagnosis is only as good as understanding how the test works and what “ingredients” went into the test to draw such a conclusion.

6.15.1 Revisiting What “Satisfying Assumptions” Means: A Brief Discussion and Suggestion of How to Approach the Decision Regarding Nonparametrics

Before we survey a couple of nonparametric tests in Python, it behooves us to comment a bit further on how we can learn whether assumptions might be satisfied for a particular model. Recall that assumptions of normality, for instance, typically involve those of the **population data** as well as the **sampling distributions** for the test statistic. How can we know whether such assumptions are satisfied? Both normality of distributions and that of sampling distributions can be considered satisfied more or less regardless of what you are seeing in your sample (not “sampling”) distributions **if you have reasonable evidence or background knowledge to assume your data were sampled from normal distributions**. For instance, if we sample IQ scores, obtain 25 scores, and are quite confident from prior research or well-known population data that IQ scores in the population are **generally normally distributed**, then even if we observe some abnormality in our limited sample, it does not necessarily mean the assumption of normality needs to be outright rejected. It may simply mean by chance we got a few scores pilling up here or there enough to generate some skew. Now you might ask, “Isn’t this what the normality tests surveyed previously in the book are for, to rule out this probability?” and the answer is, they can help, but they are not the end all and be all of things. For one, those tests (Shapiro, etc.) are relatively sensitive to assumptions themselves, and also sample size, so what they report in terms of p -values can change depending on their own deviations from assumptions. Therefore, you should not take those tests as the final verdict, which is one reason why **graphical methods** are often recommended, even by the best of data analysts, instead of relying on such tests. Usually by looking at plots that we generated in previous chapters (e.g. histograms, boxplots, etc.), it usually gives us sufficient enough of an idea regarding assumptions to know whether or not they might be tenable. Inferential tests are usually not even required.

6.15.2 Your Experience in the Area Counts

In deciding whether assumptions are tenable, the question in part also comes down to your own expertise with the material you are working with. When you look at your sample of scores, do you think “Yes, I expected to see this” or do you think “Hmm, that’s a bit high on that end of the distribution, no real reason why that should be happening though.” If it is the former, then you may be dealing with **naturally or inherently skewed distributions** to begin with, which might suggest what you are measuring has likewise an inherently skewed population distribution underlying it. However, if it is the latter, you can probably attribute such deviation to simply “randomness.” That is, you may tentatively conclude that you are seeing a bit of a “blip” in the graph in the sample data, but that you have no reason to suspect your data was drawn from a severely naturally occurring **abnormal** (i.e. in the sense of “non-normal”) distribution, given what you know about data of this kind in general. That is, why you are seeing “blips” in data becomes important because what you are witnessing in your data, if your data was collected in a sound manner, should be a very small fraction of the actual population distribution, and the assumption for the statistical test is that the population distribution is normal, not the sample data. The sample data

is just a potential “sign” that something might be amiss or not amiss. **Sample data should be interpreted with this perspective and context in mind. The particular sample data you achieved should never be considered the entire “be all” and “end all” of things.**

Hence, unless you can think of some specific reason why the sample is abnormal, you may be fine with just attributing it to general variation, and conclude something such as, “It can reasonably be assumed that such data arose from an approximately (emphasis on **approximately**) normal population based on past behavior of similar data” and carry on. Again, you might think of it this way; your sample data is 0.00001%, suppose, of all data that has been collected on similar objects. Most has already been collected (again, I am just ball-parking for demonstration). If most data over the past few years suggest normality or something close to it, even if your particular sample does not quite as much, **and if you have no specific substantive or scientific reasons to think of why normality might not hold in your sample**, then regardless of the aforementioned inferential tests and plots in this case (so long as they are not too extreme), you are probably fine in most cases with the decision to **not** reject normality. That is, if you can assume you are sampling from even approximate normal populations, then the sample distribution you obtained becomes but one “input” to add to that prior knowledge (just like my IQ distribution can supplement my prior already known info on IQs in the population). Again, the sample data is just one indicator of the underlying population distribution; there is no need to take it as the final verdict in totality.

6.15.3 What If Assumptions Are Truly Violated?

To carry our discussion further, let us assume for a moment the assumption is not satisfied and that “blip” (and blips like it in other graphs) truly suggests a skewed distribution in the population. Furthermore, the inferential tests you ran are trying to alert you to this. And, assume, as a scientist, such abnormality does not surprise you either. Let us assume you have heavy skew and you can confidently declare it to be a “real” violation. Now what? If the data are naturally skewed to a significant degree, it could suggest using an entirely different model (e.g. sometimes fitting a Poisson model or other **generalized linear model** works best with highly skewed data, especially if all distributions are coming out with similar skew shapes and it is blatantly clear the phenomena under evaluation is “naturally skewed.”). However, if most of your distributions are at least approximately normal or at minimum rectangular-like, that is probably not what is going on here. Some would say you should start performing transformations on the variable to bring it more in line, like taking square roots and such. As an example, consider numbers 4, 9, 16, 25, 49. Notice if we take square roots, we get 2, 3, 4, 5, 7. The distribution is a bit tighter. The 49 (7) is not as “extreme” from the 25 (5) anymore. Logarithms and other transformations can sometimes help as well. However, some (such as myself!) are not a big fan of this approach (i.e. that of generally transforming data). For one, transforming data is meant to **appease the statistical test** so that F -ratio inferences, etc., are more valid and that you are not committing more or less type I errors than you think you are when you set your sig. level (e.g. 0.05 or whichever you are choosing). However, we already know (or should know) p -values are not the end-all and be-all of things as well, and small or large

p -values can happen due to sample size, variability, etc., and not just a violation of assumptions. Hence, to all of a sudden be that concerned with the p -value being a bit higher or lower than it should be because of a potential assumption violation seems a bit “overkill.” What is the solution then? Do as you normally do when interpreting statistical tests, which is **to not take the p -value too darn seriously in the first place!** Use it as a **guide**, because you know all the things that it is a function of, not only the assumptions of the test.

Hence, when you get a p -value of say, 0.04, and you know your distributions might be a bit “off” more than you would like, incorporate that knowledge more or less mentally into your thought process and use the p -value as you should be doing anyway, as a **ballpark indicator** toward your scientific (not merely statistical) conclusion. As you should be doing anyway in any statistical test, you are going to look at **effect size** to see if there is anything of scientific value going on. Furthermore, ANOVA, for example, is quite **robust** to violations of normality and other assumptions, so in most cases you are likely to be fine from a **scientific perspective** because you are using the stats as a **tool** (only) to inquire about a potential scientific effect. That is, you are not trying to “appease” the aesthetics of the statistics and make everything perfectly “formal” and correct. That will never be the case anyway in any model-fitting because assumptions will always typically be “off” to some degree. It is always messy to some extent (as COVID predictions were during the pandemic, recall how most point predictions about deaths and such were way off). Incidentally, this is in part the spirit that has been promoting the “data science” and such movements, to be a bit less concerned about parametric assumptions and instead more concerned with just getting a sense of what the data have to say. In a very real way, the movement away from traditional statistical inference has been to emphasize **effects over formal inference**. That does not in itself obliterate formal statistical inference and p -values, it simply sees them for what they are (which good methodologists have been telling us for years anyway).

Check assumptions just as much for the **science** as for the **stats**. If you come across something “weird” in distribution, that may be even more important than just making sure it conforms to assumptions. **Evaluate assumptions because you want to get to know your data, first and foremost, on a scientific level.** Assumption checking might actually suggest new lines of research, you never know, because you have spotted something of scientific value. Follow a similar approach regarding spotting **outliers**. All a statistic can do is tell you if it is extreme statistically. You have to incorporate your knowledge of the material under test to know just how “extreme” it is, since you usually have only a very small fraction of the entire population, and sampling, even if done randomly with the best of intentions, does not guarantee adequate representation of a population. It is just the best we can usually do to allow chance to govern discrepancies we observe.

Also, if you have outliers in your plots (e.g. extremes in boxplots), you may have **substantive reasons** as to why a few of your points are extreme. If you do, then you may want to look at those observations more closely to make sure nothing abnormal was going on with your research units such that they violated expectations for the experiment. You could delete this or that observation, redo your analyses, and observe any difference in findings, but for most data sets, it is probably not worth doing this. Especially with a relatively small sample size on high-variability units, we usually

expect more or less messy data anyway. Now if you have 10,000 observations, and two or three are beyond z -scores of, say, 3 (or similar), then they might be worth looking at. However, if you have many “outliers” in a much smaller sample, you might start to question whether they are actually outliers, or just “data.” Case in point, the number 10 is an outlier in a data set of observations 2, 3, 10, but it is probably not an outlier from a scientific perspective – it is probably just **data**. When the next few data points come in, they might be 12 and 15, making 10 quite ordinary and normal in the updated data set. **Think about your data, do not just look at statistical indicators.** Never be too hasty to consider an innocent data point an outlier and not “belonging” to your data set. **It likely does belong. After all, it was a recorded data point like any other.** If the data recording mechanism was working properly, then you have even less justification for removing it from your data set. Again, the task is not to appease your statistical model; the task is to learn about the objects you are modeling.

So, at long last (sigh!) should you use a nonpar test for your data? It definitely would not hurt to see if you get similar p -values, and if the same decision on the null is indicated by both parametric and nonpar tests, you might report the parametric, but also note that nonpar tests did not disagree with the conclusion. In that way, if in reality the nonpar test is the most theoretically appropriate test you should be doing (i.e. in the worst case scenario your distributions are naturally skewed in the population), then at least you are covering your bases. I personally do not see a problem with this approach (some sticklers might insist you should do one test or the other and not both, but in the event of any assumption doubt, trying both I do not see as a problem). If you reject using parametric, but not with nonpar, you may simply report that finding for both (and for your own benefit, study what went into that nonpar structurally, e.g. understand how it utilized ranking and such so that you know for yourself what the test is assessing and how it is different from the parametric). Then, after reporting both, the reader can decide which results they wish to interpret.

But again, to reiterate, the focus should be on the **scientific conclusion**, not the **statistical** here. If the treatment for COVID is working (analogy again), and working well, whether a parametric or nonpar test is used **should not make that much of a difference**, and if it does (i.e. if the effect is typically not that large), reporting both gives your reader information they need to know in evaluating the result for themselves. Also, sometimes the skewness can simply suggest you still should conduct the parametric test, but also supplement your findings with things like **medians** as well as means, so readers can see for themselves how much that extended end of the distribution (for instance) is pulling the mean up or down. That is not changing your model necessarily, it is rather addressing characteristics of your data by reporting less “sensitive” measures that are not as influenced by extremes. Range statistics might be a nice idea (such as the semi-interquartile range) to go along with the median as a more “resistant” or “robust” measure of dispersion, since the standard deviation will be influenced by the degree of skew as well. So in this approach, you are reporting parametric tests, but supplementing them with more resistant summary measures to account for skew, informing your readers of the assumption “imperfection,” and switching the issue back to the scientific problem and interpretation of data where it belongs, rather than over-reliance on p -values.

Given this discussion, if you still do decide a nonpar test is useful, we now consider a couple of the more popular options. For details on other tests, see Howell (2002) for

a succinct summary or Siegel and Castellan (1988) for a more thorough (and excellent) treatment.

6.15.4 Mann-Whitney U Test

When the assumptions of the independent samples t -test are violated or unverifiable, the **Mann-Whitney U test** becomes a viable alternative nonpar option. Recall that the null hypothesis in the independent samples t -test was that both population means were equal. That is, the null hypothesis was

$$H_0 : \mu_1 = \mu_2$$

Given a rejection of the null, the alternative $H_1 : \mu_1 \neq \mu_2$ was inferred. The Mann-Whitney U test evaluates the null that both samples were drawn from the same population, but it does so based on the **ranks of the data** instead of their continuous scores. Hence, instead of focusing on score differences such as that between 8 and 10, the Mann-Whitney test emphasizes ranking differences, where 10 may be “first” and 8, “second.” This naturally implies that if the data cannot be ranked in some capacity, then the Mann-Whitney test cannot be used. When is data not “rankable?” Nominal data, recall by definition, cannot be ranked. For example, if your favorite color is blue and mine red, the variable “favorite color” is not being measured on a scale that has a natural ranking. Now, that does not imply the colors could not be ranked by some other measure than favorite color. Indeed, if we examined the physical qualities of color, we may possibly rank them on measures of hue or saturation. The point is, however, is that if the scale you are using is not one with at least a natural ranking quality, then it implies that you are dealing with a nominal scale, and the Mann-Whitney test or other similar ranking tests would make no sense to perform. Hence, data for the test should be **at least at the level of ordinal** such that magnitudes between measured quantities can be inferred and make good sense to infer them.

Be sure to note that your data do not need to be merely at the level of ordinal. They can be interval or ratio, since these scales can be **collapsed** into ordinal scales for the purpose of the test. That is, pseudo-continuity can easily be collapsed into a ranking, but not necessarily vice versa. The point is, however, that your data cannot be “less” than ordinal and has to be **at least ordinal** for the test to make any sense performing.

We demonstrate the test in Python. For this example, we will use the **grade data**, on which grades of passing (1) or failing (0) were recorded along with the amount of time studied for a test (recorded in minutes). We first build our data set:

```
data = {'grade':[0, 0, 0, 0, 0, 1, 1, 1, 1, 1],
        'studytime':[30, 25, 59, 42, 31, 140, 90, 95, 170, 120]}
df = pd.DataFrame(data)
df
Out[75]:
   grade  studytime
0      0         30
1      0         25
2      0         59
3      0         42
```



```

4      0      31
5      1     140
6      1      90
7      1      95
8      1     170
9      1     120

```

We next pull out and identify the columns of `grade` and `studytime` from the data-frame `df`:

```

grade = df['grade']
studytime = df['studytime']

```

We import **scipy** to conduct the test, on which we will use `stats.mannwhitneyu()`:

```

import scipy
from scipy.stats import mannwhitneyu

Out[269]: <function scipy.stats.stats.mannwhitneyu(x, y, use_
continuity=True, alternative=None)>

```

We now define our samples and then conduct the test:

```

sample1 = 30, 25, 59, 42, 31
sample2 = 140, 90, 95, 170, 120
stat, p = mannwhitneyu(sample1, sample2)
stat, p

Out[514]: (0.0, 0.006092890177672406)

```

We see that the result is statistically significant with a p -value equal to approximately 0.006. Hence, we can reject the null and have evidence to suggest that the two groups were not drawn from the same population. The Mann-Whitney test is quick, easy, and immediately gives you an idea as to whether something might be “going on” with your data without having to satisfy parametric assumptions.

6.15.5 Kruskal-Wallis Test as a Nonparametric Alternative to ANOVA

Recall that when we have several means to compare, conducting numerous independent samples t -tests unduly inflates the type I error rate over the family of comparisons. The solution in such a case is the one-way ANOVA. When assumptions such as normality and equal variances are violated, a plausible nonpar alternative is the Kruskal-Wallis (K-W) test to evaluate the null that all samples were drawn from the same population. To demonstrate the test in Python, we recall the achievement data on which we computed an ANOVA earlier in the book. For this analysis, we organize each sample into its own vector. Recall that there were four samples and hence we have four data vectors:

```

x = [70, 67, 65, 75, 76, 73]
y = [69, 68, 70, 76, 77, 75]
z = [85, 86, 85, 76, 75, 73]

```

```
w = [95, 94, 89, 94, 93, 91]
```

We use `stats.kruskal()` to run the test:

```
from scipy import stats
stats.kruskal(x, y, z, w)
```

```
Out[127]: KruskalResult(statistic=16.26653554778557, pval-
ue=0.0009998585717080008)
```

We can see that the p -value for the test is quite small at a value of 0.0009, certainly smaller than some pre-set level such as 0.05. Hence, we reject the null that the samples were drawn from the same population and infer that they were not. However, as with ANOVA, simply because we have rejected the overall null, we do not yet know where the differences may lie. To know this, we need to conduct post-hoc tests to determine which populations are pairwise different from which others. One easy way to do this is to simply run the K-W test on pairwise groups and issue a correction on alpha for each comparison. For example, we will see if there is a difference between groups x and y :

```
stats.kruskal(x, y)
Out[128]: KruskalResult(statistic=0.5247349823321599, pval-
ue=0.46882883249379537)
```

We can see that the test comes out to be non-statistically significant ($p = 0.4688$), suggesting that a difference does not exist between these two groups. We could do the same thing for the other groups as well to determine where differences may lie, testing each comparison at a smaller significance level to keep the overall family-wise error rate low. An even more effective strategy would be to use the **Nemenyi test** as a post-hoc test, available in the package **scikit-posthocs**:

```
pip install scikit-posthocs
import scikit_posthocs as sp

v = [[70, 67, 65, 75, 76, 73], [69, 68, 70, 76, 77, 75], [85, 86,
85, 76, 75, 73], [95, 94, 89, 94, 93, 91]]
sp.posthoc_nemenyi(v)

Out[283]:
```

	1	2	3	4
1	-1.000000	0.974034	0.387602	0.003552
2	0.974034	-1.000000	0.657109	0.016002
3	0.387602	0.657109	-1.000000	0.286205
4	0.003552	0.016002	0.286205	-1.000000

We can see that the test reports significant differences between groups 1 and 4 ($p = 0.003552$), as well as 2 and 4 ($p = 0.016002$).

Review Exercises

1. Explain how the **t-test** can be considered a special case of the wider **analysis of variance model**.
2. Why is conducting **multiple t-tests** on the same data set a bad idea to discern pairwise mean differences? What advantage does the ANOVA afford over this? Explain.
3. Consider the **pooled variance** term in the *t*-test:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Show that when sample size is equal per group, that is, $n_1 = n_2$, whether one uses a pooled term or the original estimate of variance for the two-sample case, will amount to the same thing.

4. What is the nature of a **Bonferroni** adjustment? If one has only two means to compare, is doing a Bonferroni adjustment useful? What is the drawback of performing a Bonferroni adjustment when one is working with several means?
5. Explain to someone the **logic of ANOVA** from the ground up and from first principles. Start with a data layout and explain exactly what ANOVA seeks to do.
6. R.A. Fisher, innovator of the analysis of variance, once said that ANOVA was nothing more than an exercise in “**arranging the arithmetic**” and nothing beyond that. From a scientific point of view, explain what Fisher might have meant by this statement. What does this mean to you, as a scientist?
7. Discuss what information **eta-squared**, η^2 , provides that an *F*-statistic in ANOVA does not. What is the potential advantage of interpreting η^2 in conjunction with *F*? How about interpreting it **instead** of *F*? Would that be a good idea?
8. Discuss the nature of a **post-hoc test**. What is the purpose of a post-hoc test, and how can it be that different post-hoc tests could yield different conclusions? Isn’t “a mean difference a mean difference?” How could using a different post-hoc test ever **not** produce a difference? Explain.
9. A researcher decides to perform a **two-way ANOVA** without interaction instead of two one-way ANOVAs. What are some implications for this in terms of interpreting results?
10. What is the purpose of performing a **factorial ANOVA** over two one-way ANOVAs?
11. Your colleague says to you, “In virtually every factorial ANOVA conducted on the planet, there is an **interaction in the sample**, but not necessarily in the **population** from which the data were drawn.” Is your colleague correct? Defend or correct him.
12. Run an ANOVA on the **iris** data where sepal length is the DV and species the IV. Evaluate the null hypothesis that mean sepal length is the same across species. If you reject, compute a measure of effect size.
13. Discuss the difference between a **parametric** and a **nonpar** test. What is the rationale for performing a nonpar test and how should this decision be approached? Explain.